Band-1 Sunflower Bound: Next Log-Factor Probe (Collapse List 1.22)

Christopher Baird + ZoraASI Band-1 dash (executor)

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Honesty stamp. ALWZ opened the modern sunflower line; subsequent improvements are **incremental**. This note does **not** claim that the Erdős–Rado sunflower conjecture is solved. **GATE-04** is locked out of scope. No Clay / ToE stamp.

> **Erdős sunflower conjecture remains OPEN.** Best *published journal* upper form is still $(Cr \log w)^w$ (BCW 2021). The live Band-1 target is to remove the remaining $\log w$ for *ordinary* sunflowers.

Abstract

Literature ledger (1960–2026 sweep) cites Erdős–Rado, ALWZ, Rao, FKNP, and Bell–Chueluecha–Warnke. **Cited best published bound:** $(Cr \log w)^w$ (BCW, *Discrete Mathematics* 344 (2021), 112367). Fukuyama arXiv:2510.19037 (sub-log base claim) is ledger-noted as **preprint only** — not elevated. Box work: unit-tested detector; classical transversal certificates through size **81**; tiny exact max sunflower-free enumerations on $n \leq 6$ — **illustration, not a new asymptotic**.

Problem (collapse 1.22)

An **r-sunflower** is r sets whose pairwise intersections equal the total intersection (core). Let $f(w, r)$ be the least m such that every family of m distinct w-sets contains an r-sunflower.

- **Erdős–Rado (1960):** $f(w, r) \leq w!(r-1)^w + 1$.
- **Lower bound:** $f(w, r) > (r-1)^w$ (transversal / product construction).
- **Conjecture:** for each fixed $r \geq 3$, some $C = C(r)$ with $f(w, r) \leq C^w$.
- **Live incremental target:** pass from $(Cr \log w)^w$ to $C(r)^w$ for ordinary sunflowers.

Literature ledger (2024–2026 status; cite carefully)

Year	Who	Bound form (schematic)	Status
1960	Erdős–Rado	$w!(r-1)^w + 1$	classical; conjecture posed
1990s	Kostochka	modest $r = 3$ improvements	pre-breakthrough

Year	Who	Bound form (schematic)	Status
2019/2021	**ALWZ**	$(Cr^3 \log w \log\log w)^w$	**opened** modern line (STOC 2020; *Annals* 194 (2021), 795–815; arXiv:1908.08483). Log necessary for *robust* sunflowers
2020	Rao	$(Cr \log(rw))^w$	*Discrete Analysis* 2020:2
2020	Tao	Rao-scale entropy re-proof	blog exposition
2020	FKNP	spread / expectation-threshold	feeds Rao sharpening
2021	**Bell–Chueluecha–Warnke**	$(Cr \log w)^w$	**BEST PUBLISHED JOURNAL FORM** — *Discrete Math.* 344 (2021), 112367; DOI 10.1016/j.disc.2021.112367
2023	Rao survey	BCW cited as SOTA	*Bull. AMS* 60 (2023), 29–38
2025	Fukuyama	claimed $(ck^2 \ln m / \ln\ln m)^m$	arXiv:2510.19037 v2 — **PREPRINT ONLY; not elevated**
2026	structural notes	special cases	e.g. arXiv:2606.02667 — do not close full conjecture

Canonical best-published citation. T. Bell, S. Chueluecha, L. Warnke, *Note on sunflowers*, Discrete Mathematics **344** (2021), 112367.

Hygiene. Sub-log / C^w preprint claims after BCW are ledger-noted and **not** treated as published SOTA unless peer-reviewed. ALWZ already showed a log is necessary for *robust* sunflowers; whether ordinary sunflowers need $\log w$ remains the open incremental question.

Method (box) — illustration only

1. **Detector.** Brute-force over $\binom{|F|}{r}$; sunflower iff all pairwise intersections equal $\bigcap$ of all members.
2. **Transversal re-check.** Build size-$(r-1)^w$ product family; certify r-sunflower-free.
3. **Tiny exact max free.** Branch-and-bound largest r-sunflower-free w-uniform family on universe $[n]$ for $n \le 6$.
4. **Gap proxy.** Illustrative $C = 1$ envelopes comparing BCW form to conjectured r^w (functional forms only).

Results

Unit tests

`PASS`: empty-core matching sunflower; K_3 edges not a sunflower; star sunflower; transversal $(w, r) = (3, 3)$ size 8 free; exact max free on $(n, w, r) = (4, 2, 3)$ equals **4**.

Transversal certificates (classical lower construction)

w	r	$\|F\| = (r-1)^w$	r-sunflower?	Time (s)
2	3	4	no	6×10^{-6}
3	3	8	no	6×10^{-5}
3	4	27	no	0.020
4	3	16	no	6×10^{-4}
4	4	**81**	**no**	**1.75**
5	3	32	no	0.006

Tiny exact max sunflower-free families (illustration)

n	w	r	$\binom{n}{w}$	max free	Time (s)
4	2	3	6	**4**	$< 10^{-4}$
5	2	3	10	**5**	0.001
6	2	3	15	**6**	0.013
5	3	3	10	**6**	0.001
6	3	3	20	**10**	0.29

Label: exact finite combinatorics on tiny universes — **not** a new asymptotic constant and **not** a proof that the log-factor drops.

Bound-form gap proxy ($r = 3$, illustrative $C = 1$)

Under $C = 1$ normalizations the BCW/conjecture ratio is exactly $(\log w)^w$ — the **live remainder** after BCW. Published absolute C are larger; table is schematic only.

Reproducibility

- Lab: `/workspace/mission-control/experiments/band1/sunflower_bound/`
- Run: `python3 sunflower_bound_probe.py`
- Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`
- Wall-clock full probe: ≈ 2.09 s

Conclusion

Item	Status
Best published journal bound	$(Cr \log w)^w$ **(BCW 2021)**
Conjecture $C(r)^w$	**OPEN**
Box contribution	transversal re-check + tiny exact free families — illustration
Collapse 1.22	next log-factor still the incremental program
GATE-04 / Clay / ToE	out of scope / locked

Do not claim Erdős sunflower solved.